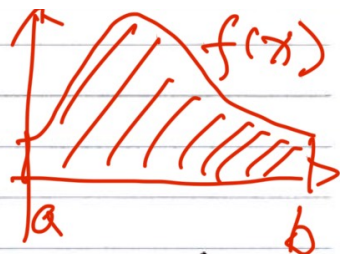


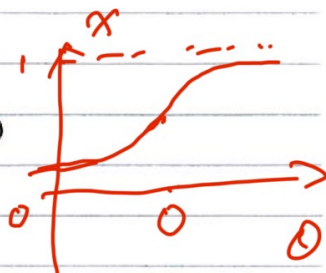
{ Numerical Quadrature :

$$A = \int_a^b f(x) dx, \quad a, b \text{ finite}$$



Note that, If, a, b not finite, we need to do a transformation.

$$\theta = t(x) = \log \frac{x}{1-x} \Rightarrow x = t^{-1}(\theta)$$

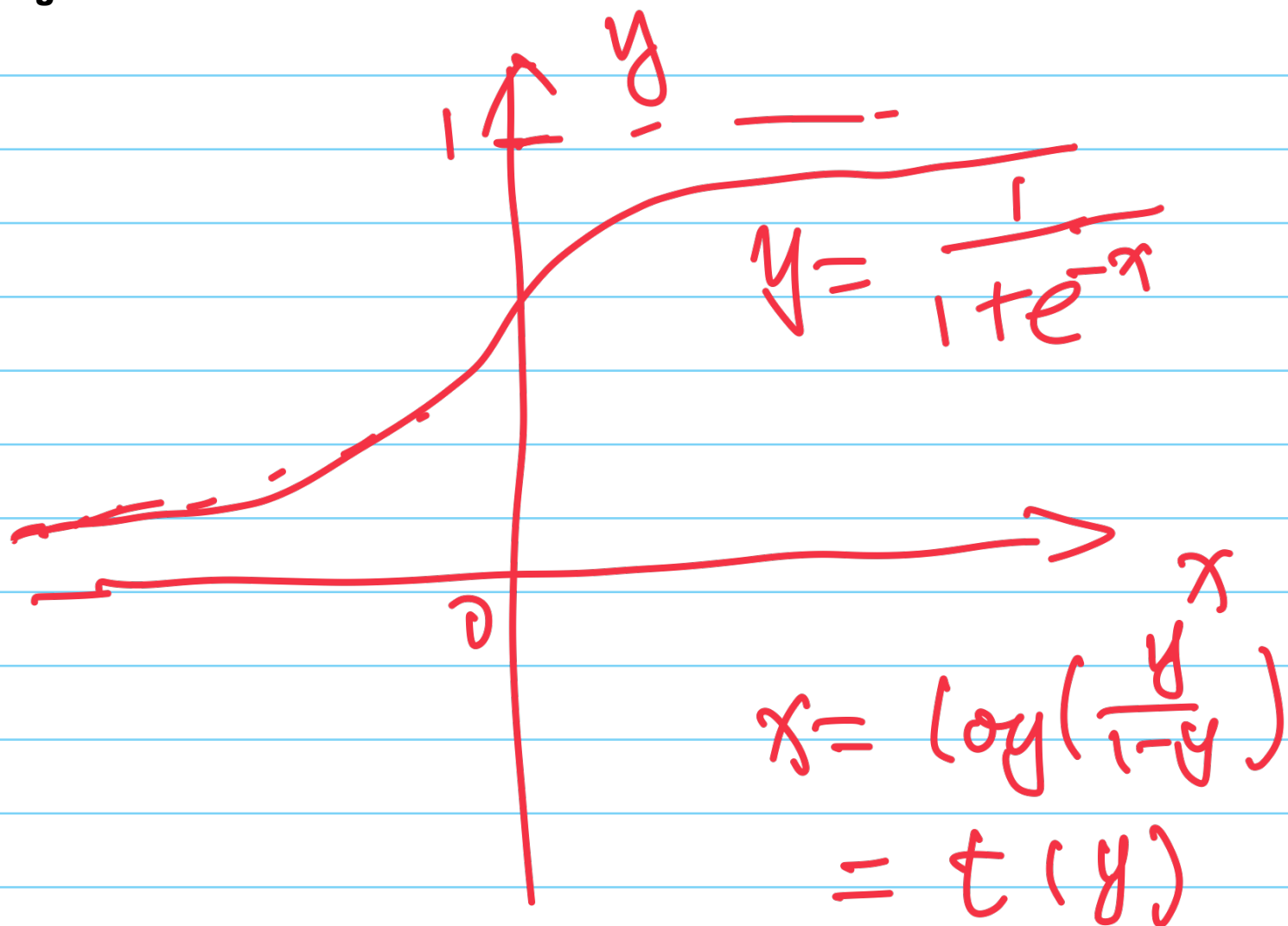


If $a = -\infty$, $b = +\infty$, then

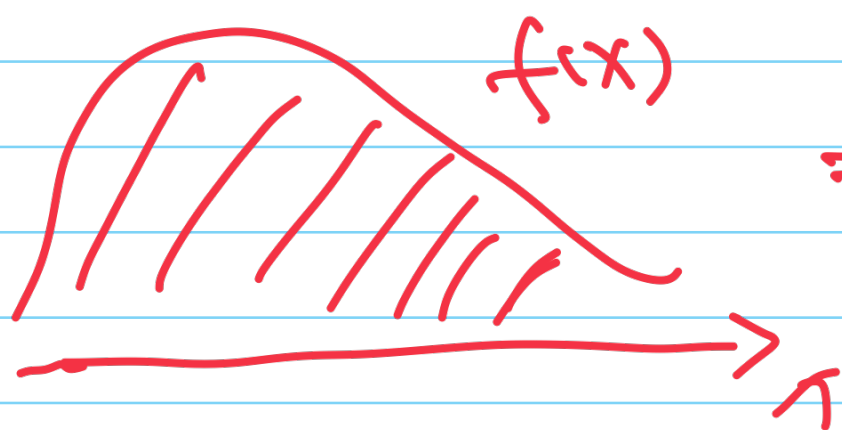
$$\int_a^b f(x) dx = \int_{-\infty}^{\infty} f(x) dx = \int_0^1 f(t^{-1}(\theta)) d t^{-1}(\theta)$$

}

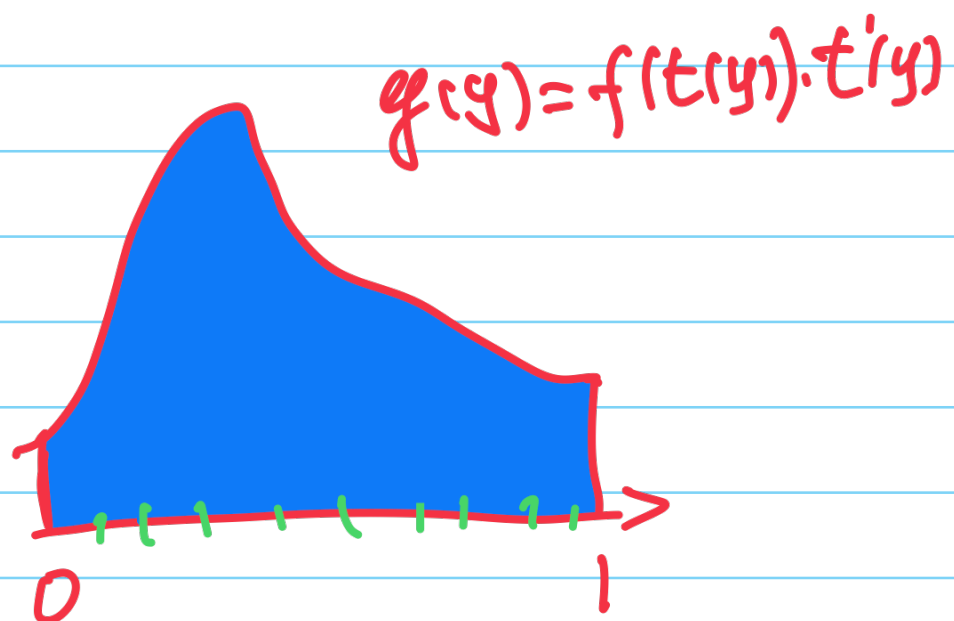
Example: Logistic Transformation



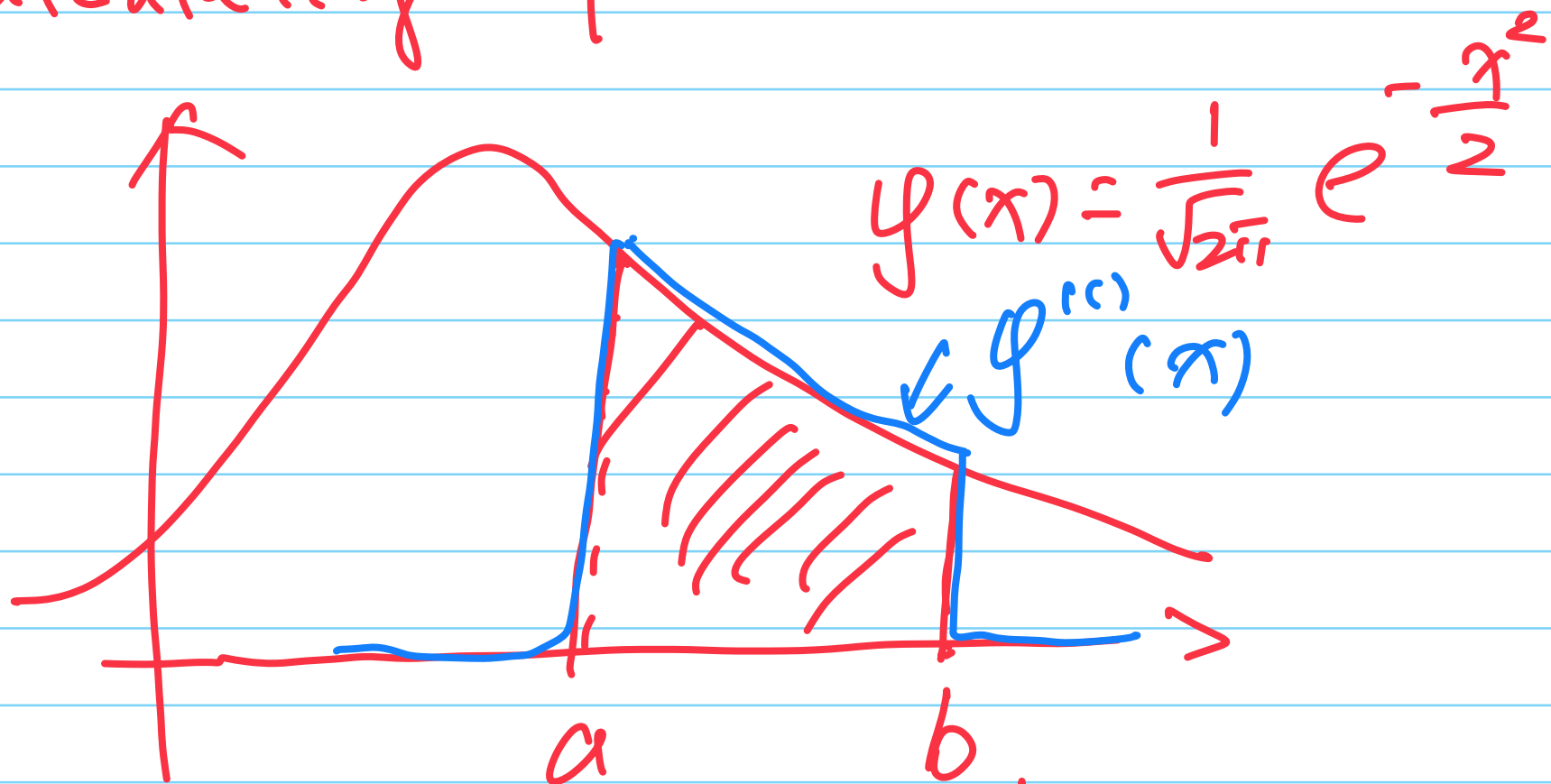
$$\int_{-\infty}^{\infty} f(x) dx = \int_0^1 f(t(y)) \cdot t'(y) \cdot dy$$



$\Rightarrow$



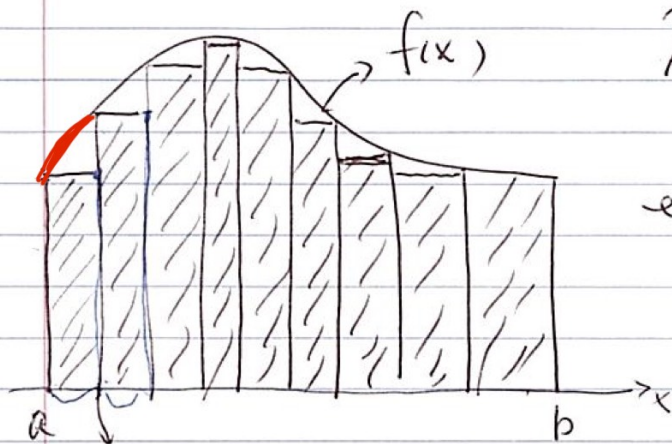
Calculating prob.



$$P(a < X < b) = \int_a^b f^{(c)}(x) dx$$

$$f^{(c)}(x) = f(x) I(a < x < b)$$

=> Riemman Rule



$$\tilde{A}_n = \sum_{i=0}^{n-1} h_n \cdot f(a + i h_n)$$

$$\text{error} = O\left(\frac{1}{n}\right)$$

$a + h_n$

$$h_n = \frac{b-a}{n}$$

⇒ Midpoint Rule

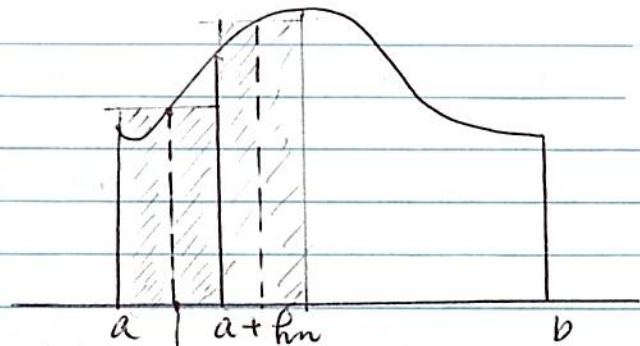
$$h_n = \frac{b-a}{n}$$

Formula:

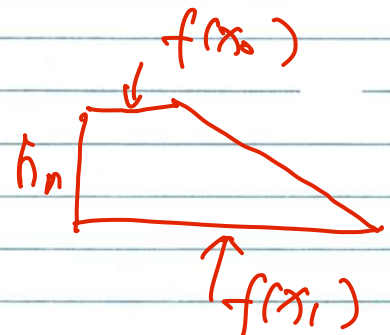
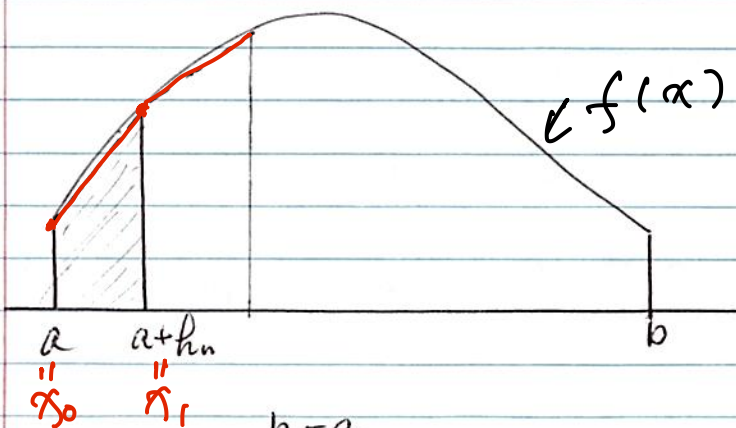
$$\tilde{A}_n = \sum_{i=1}^n h_n \cdot f\left(a + \left(i - \frac{1}{2}\right) h_n\right)$$

$\swarrow$
 $a + \frac{1}{2} h_n$

$$\text{error} = O\left(\frac{1}{n}\right)$$



⇒ Trapezoidal Rule



⇒ $h_n = \frac{b-a}{n}$

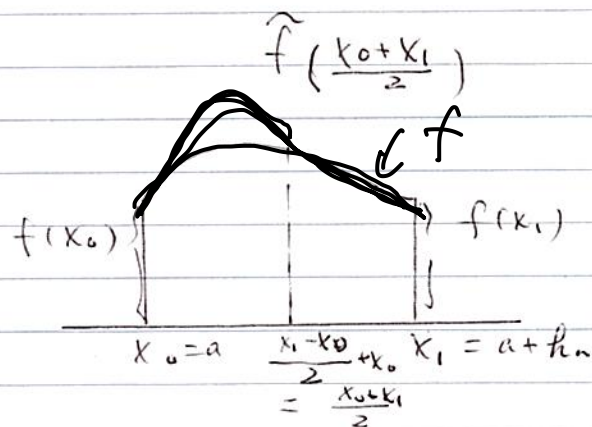
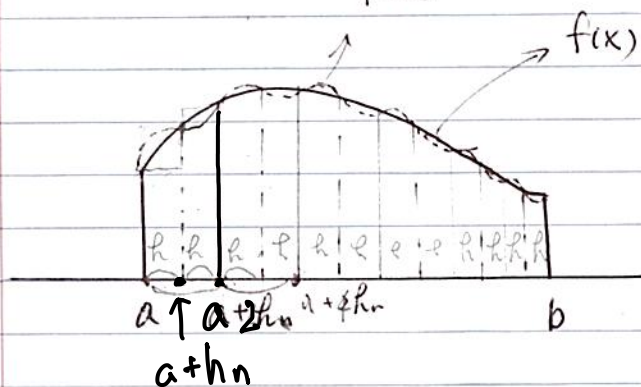
⇒ Formula: $\tilde{A}_n = \frac{1}{2} (f(x_0) + f(x_1)) \cdot (x_1 - x_0)$

$$\begin{aligned} \tilde{A}_n = \frac{1}{2} h_n [& f(a) + f(a+h_n) + f(a+h_n) + \dots \\ & + f(a+(n-1)h_n) + f(a+(n-1)h_n) \\ & + f(b)] \end{aligned}$$

$$= \frac{1}{2} h_n [f(a) + f(b)] + h_n \cdot \sum_{i=1}^{n-1} f(a + i h_n)$$

$\Rightarrow$ Simpson's Rule

$$\hat{f}(x) = f(x)$$



$$h_n = (b-a) / n \quad (\text{let } n = 2m) \quad n \text{ is even number}$$

$$\begin{aligned} \int_{x_0}^{x_1} \hat{f}(x) dx &= \frac{x_1 - x_0}{6} \left[\hat{f}(x_0) + 4 \hat{f}\left(\frac{x_0 + x_1}{2}\right) + \hat{f}(x_1) \right] \\ &= (x_1 - x_0) \left[\frac{1}{6} \hat{f}(x_0) + \frac{4}{6} \hat{f}\left(\frac{x_0 + x_1}{2}\right) + \frac{1}{6} \hat{f}(x_1) \right] \end{aligned}$$

$$\begin{aligned} \hat{A}_{2m} &= \frac{2h_n}{6} \left[f(a) + 4f(a+h_n) + f(a+2h_n) \right. \\ &\quad + f(a+2h_n) + 4f(a+3h_n) + f(a+4h_n) \\ &\quad + f(a+4h_n) + 4f(a+5h_n) + f(a+6h_n) \\ &\quad + \dots + f(b-2h_n) + 4f(b-h_n) + f(b) \left. \right] \end{aligned}$$

$$\begin{aligned} &= \frac{h_n}{3} \left[f(a) + f(b) + 4 \sum_{i=1}^m f(a + (2i-1)h_n) \right. \\ &\quad \left. + 2 \sum_{i=1}^{m-1} f(a + 2ih_n) \right] \end{aligned}$$

An example of computing marginalized likelihood

We demonstrated the midpoint rule with a simple example.

Example (Normal data with log-norm prior for σ). The data $X_1, \dots, X_n$ are modelled as IID samples from a normal distribution given the value of the mean parameter μ and the logarithm of the standard deviation σ , denoted by w . The likelihood function of μ and w based on data $X_1, \dots, X_n$ is therefore:

$$L(\mu, w) = \prod_{i=1}^n \phi(x_i; \mu, e^w)$$

where $\phi(x; \mu, \sigma)$ is the density function of Normal distribution with mean μ and σ .

We assign normal priors to parameter μ and w with fixed $\mu_0, \sigma_\mu, w_0, \sigma_w$:

$$\begin{aligned}\mu &\sim N(\mu_0, \sigma_\mu) \\ w &\sim N(w_0, \sigma_w)\end{aligned}$$

The density function of the prior for μ and w is therefore:

$$P(\mu, w) = \phi(\mu; \mu_0, \sigma_\mu) \phi(w; w_0, \sigma_w)$$

The posterior distribution of μ and w can be found using Bayes rule:

$$P(\mu, w \mid X_1, \dots, X_n) = \frac{L(\mu, w) P(\mu, w)}{\int_{\mu} \int_w L(\mu, w) P(\mu, w) d\mu dw}$$

We will want to compute the normalizing constant $\int_{\mu} \int_w L(\mu, w) P(\mu, w) d\mu dw$, which is needed to compare this model with other models. Since this integral is intractable, we will use midpoint rule to approximate it. The range of μ and w are both infinite, we therefore apply the following change of variable to μ and w before applying midpoint rule:

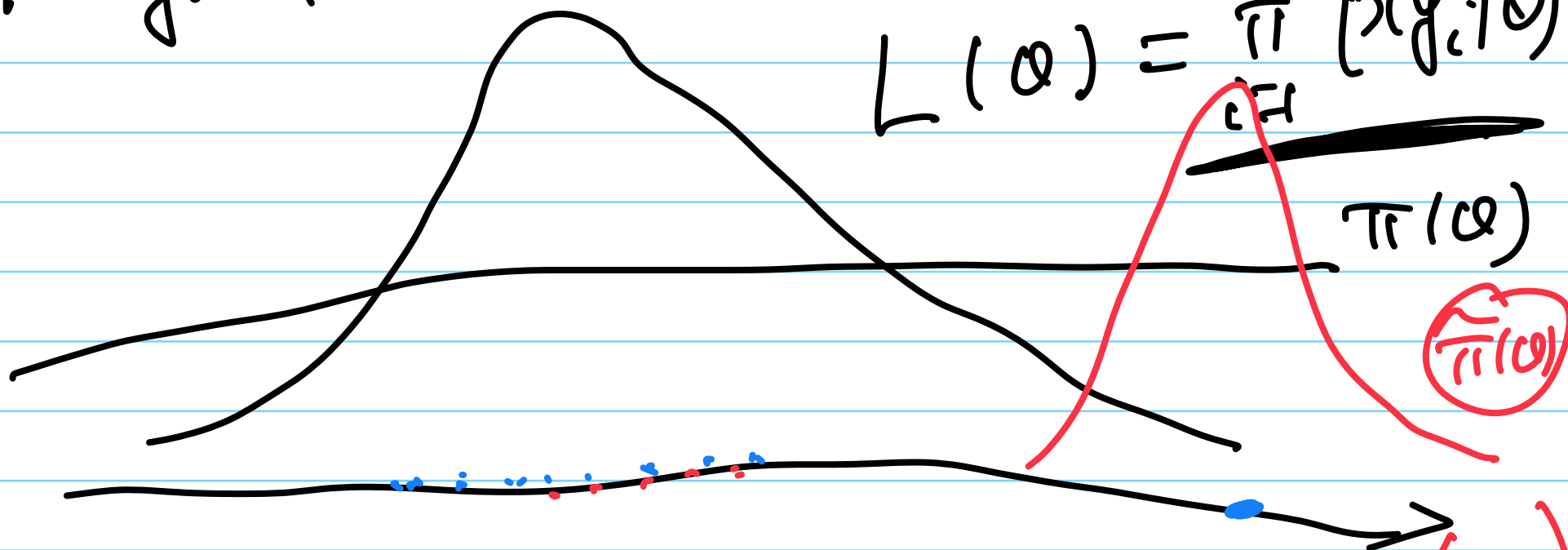
$$\begin{aligned}\mu(\mu_t) &= \log(\mu_t) - \log(1 - \mu_t) \\ w(w_t) &= \log(w_t) - \log(1 - w_t)\end{aligned}$$

We then compute the following integral for computing the normalizing constant:

$$\int_{\mu_t} \int_{w_t} L(\mu(\mu_t), w(w_t)) P(\mu(\mu_t), w(w_t)) \frac{1}{\mu_t(1 - \mu_t)} \frac{1}{w_t(1 - w_t)} d\mu_t dw_t$$

Marginal Likelihood

$$L(\theta) = \prod_{i=1}^n p(y_i | \theta)$$



$$p(y) = \int L(\theta) \cdot \pi(\theta) d\theta = E_{\pi}(L(\theta))$$

$$= \int \prod_{i=1}^n p(y_i | \theta) \cdot \pi(\theta) d\theta$$

$$p(\theta | y) = \frac{\prod_{i=1}^n p(y_i | \theta) \cdot \pi(\theta)}{p(y)}$$

Example

$$y_1, \dots, y_n \sim N(\mu, \sigma^2)$$

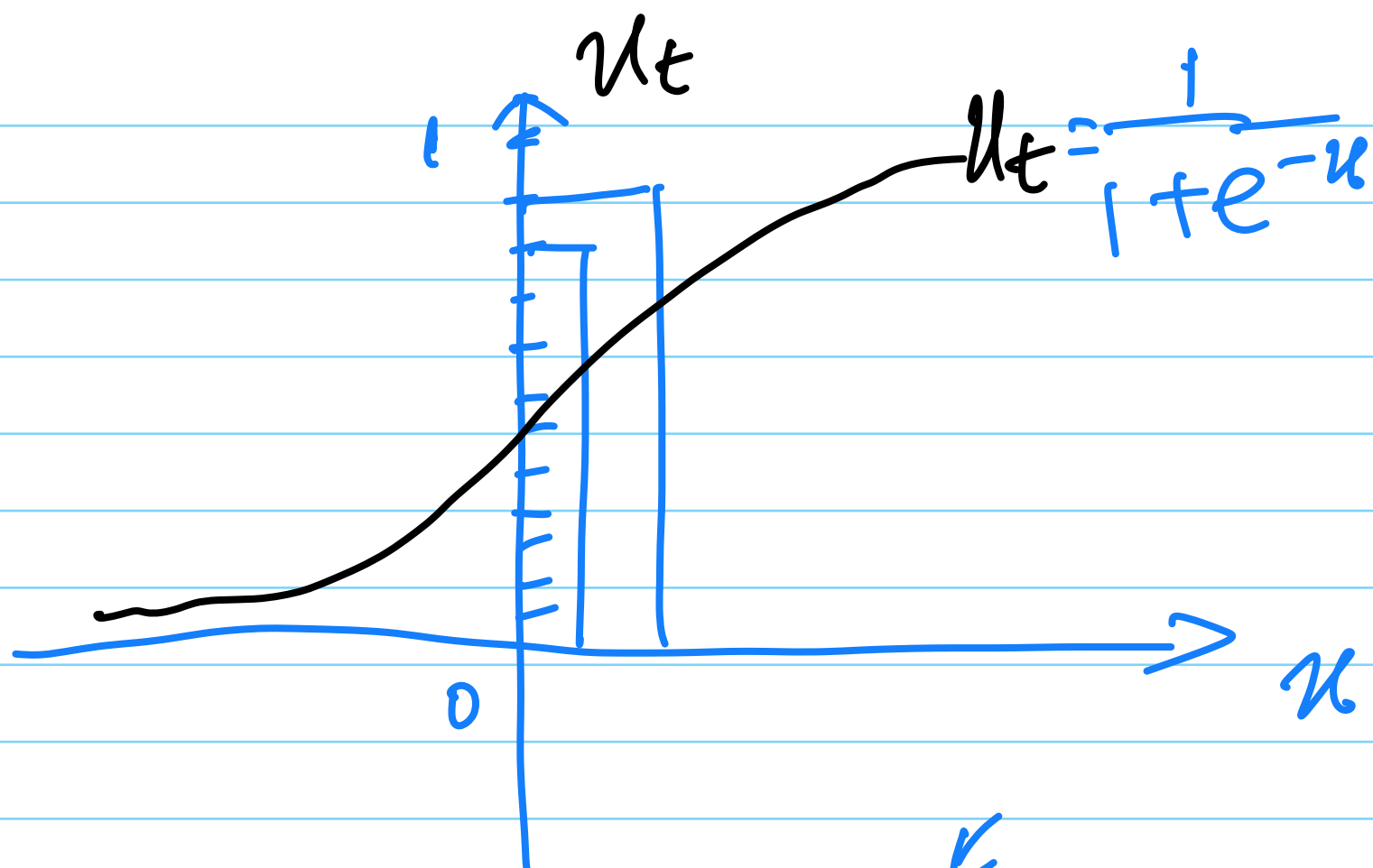
$$\mu \sim \pi_\mu(\mu), \quad \mu \sim N(\mu_0, \sigma_0^2)$$

$$\sigma^2 \sim \pi_{\sigma^2}(\sigma^2), \quad \searrow$$

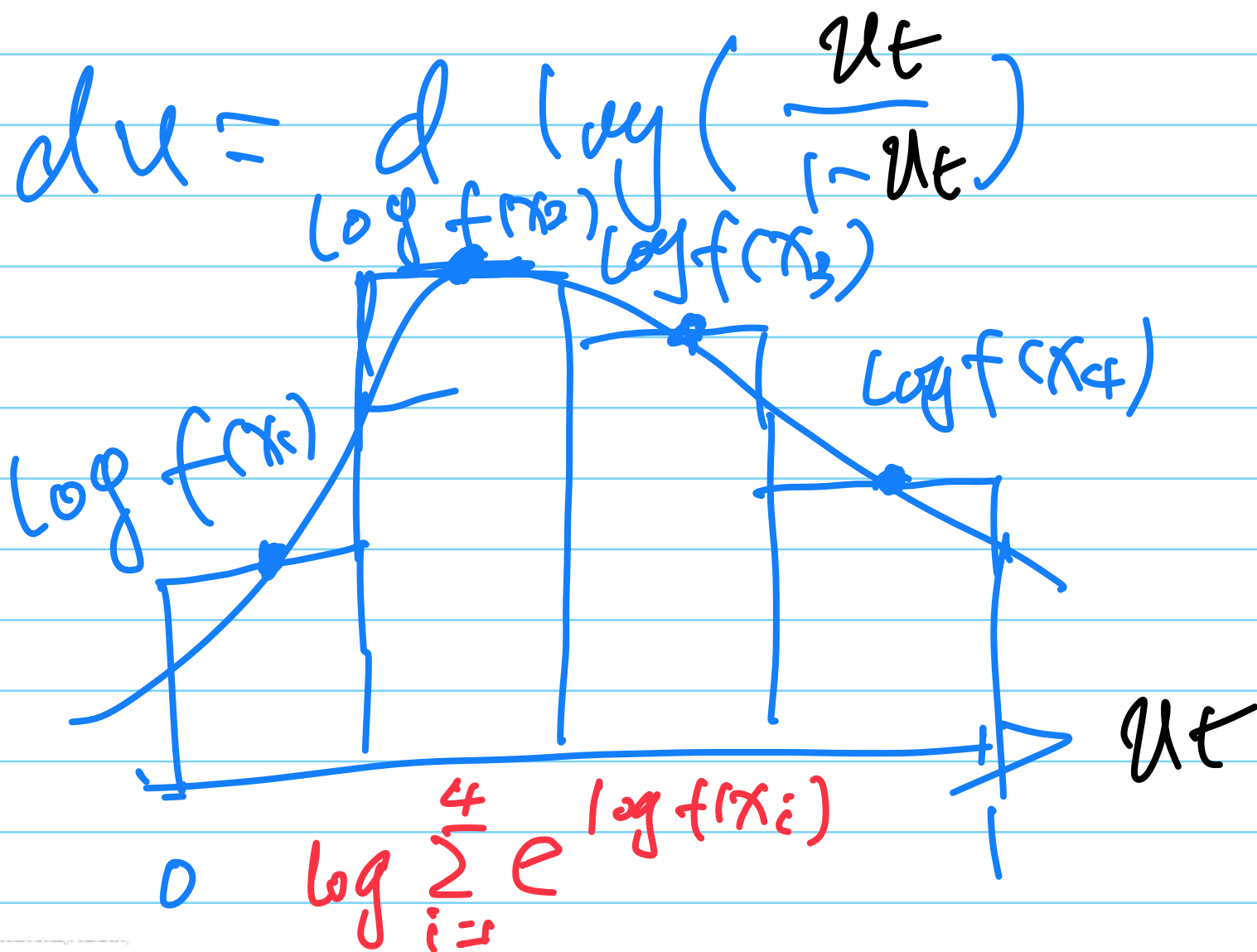
$$W = \log \sigma^2 \sim N(\mu_1, \sigma_1^2) \quad \nwarrow \pi$$

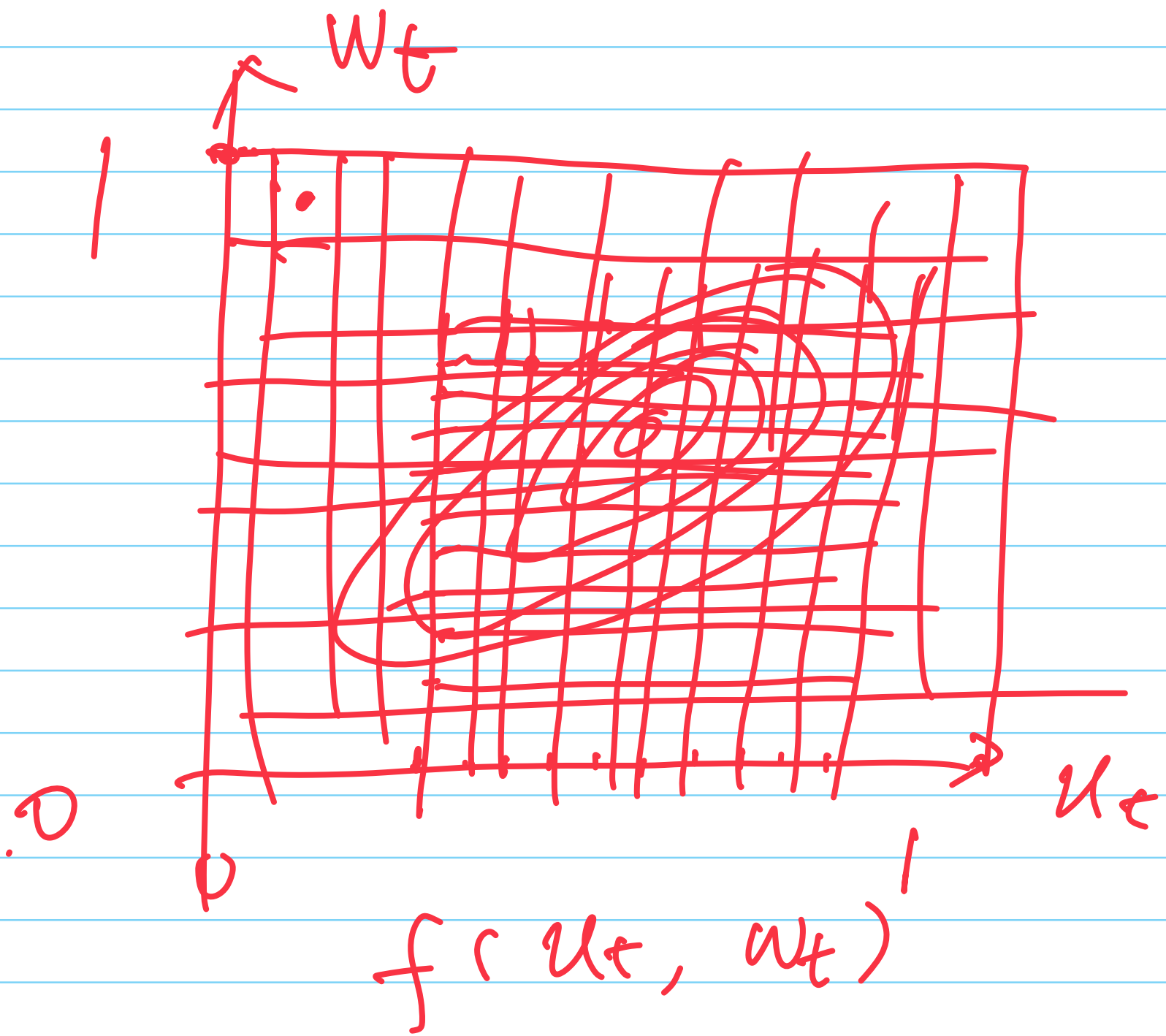


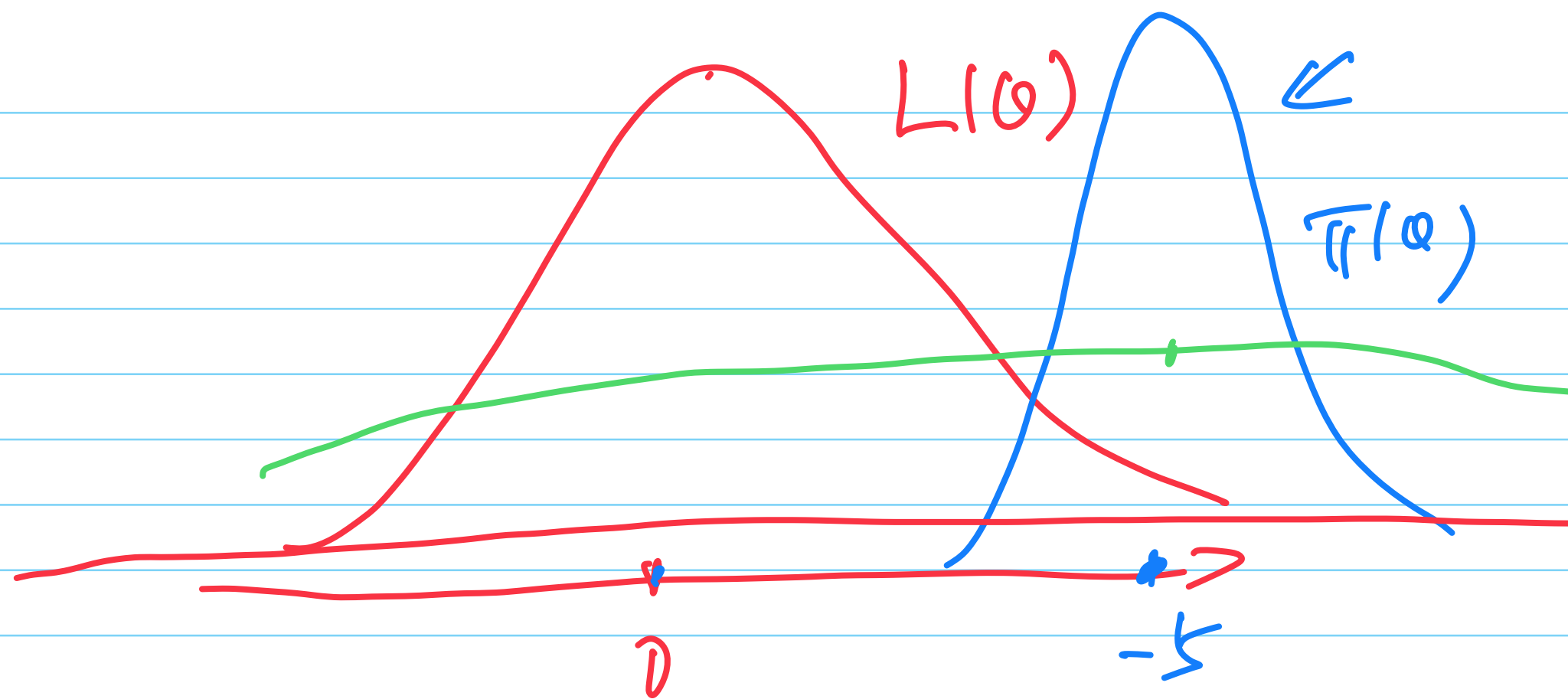
$$ML = \int \int \left(\prod_{i=1}^n P(y_i | \mu, \sigma^2) \cdot \pi(\mu) \pi(\sigma^2) \right) d\mu d\sigma^2$$
$$= \int \int \underline{f(\mu, \sigma^2)} d\mu d\sigma^2$$



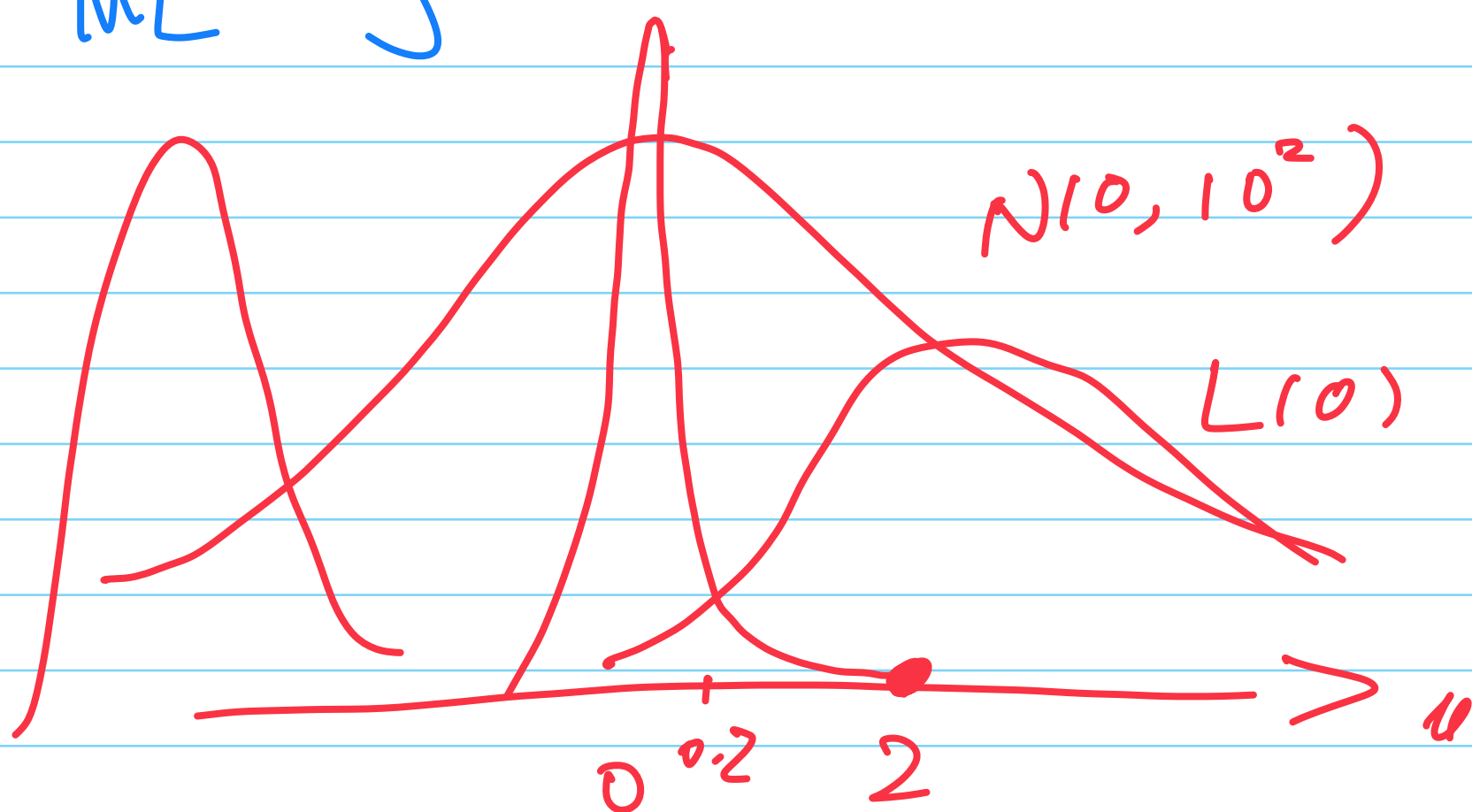
$$u = \log \left(\frac{u_t}{1 - u_t} \right)$$





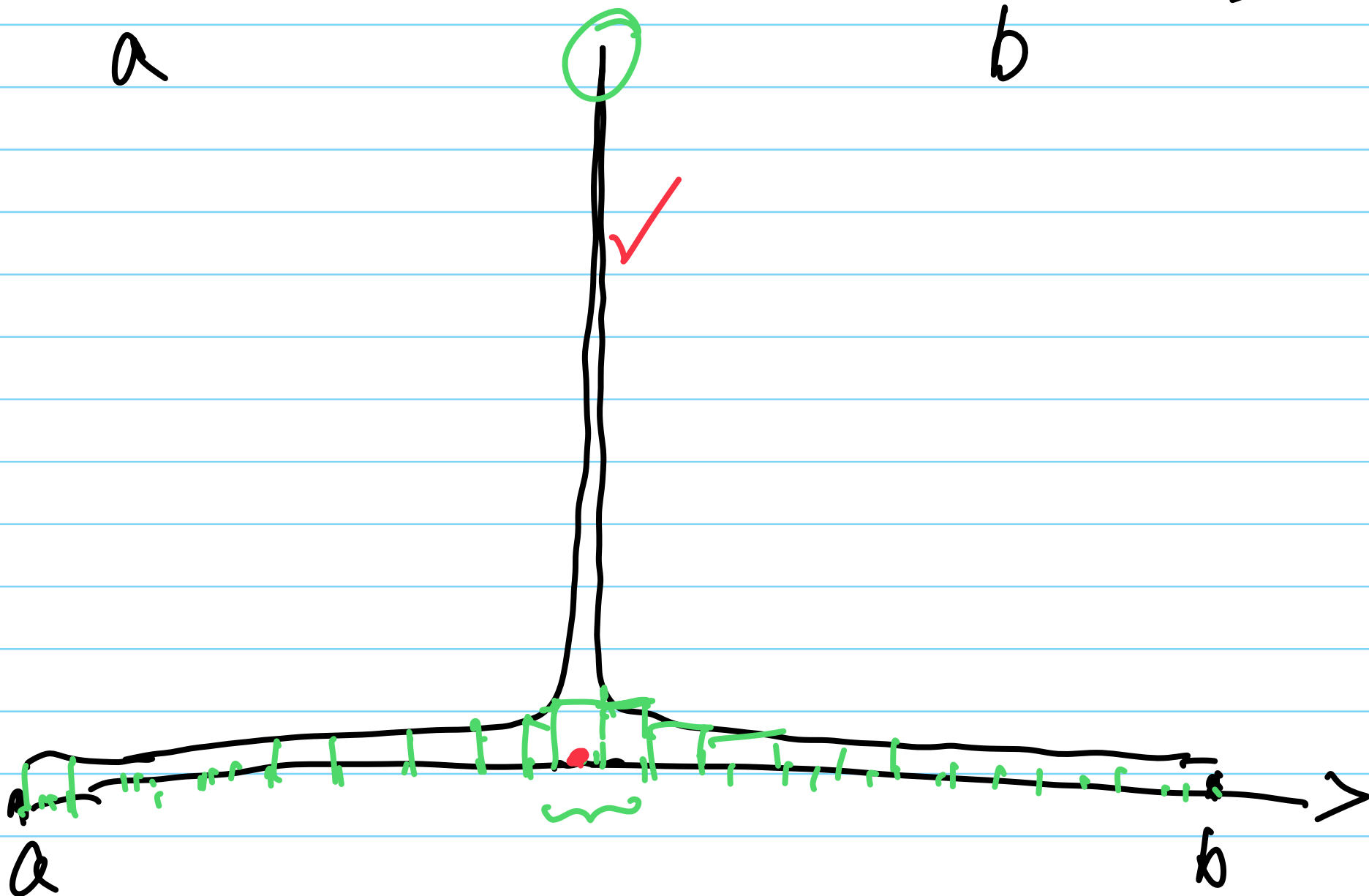
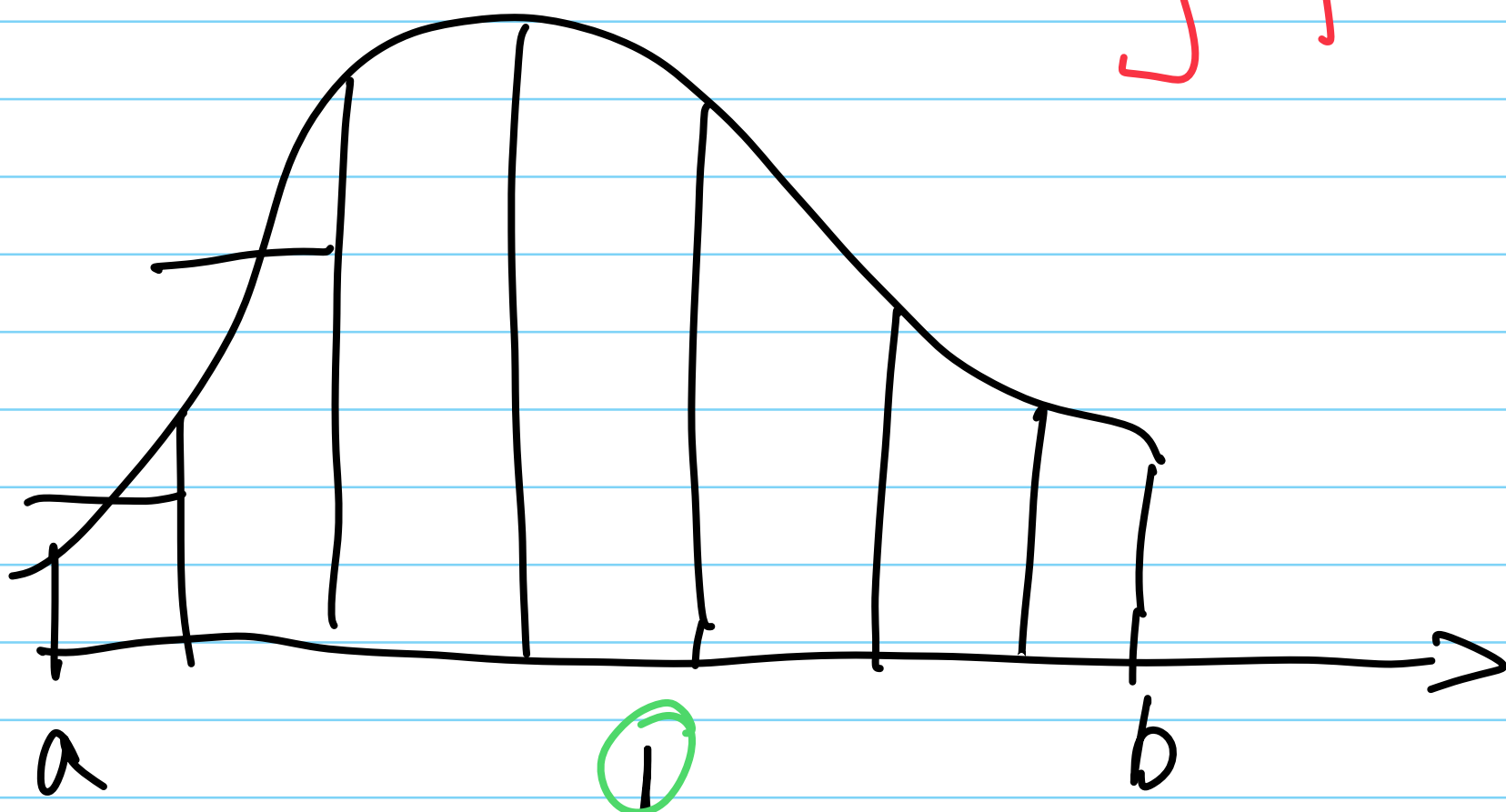


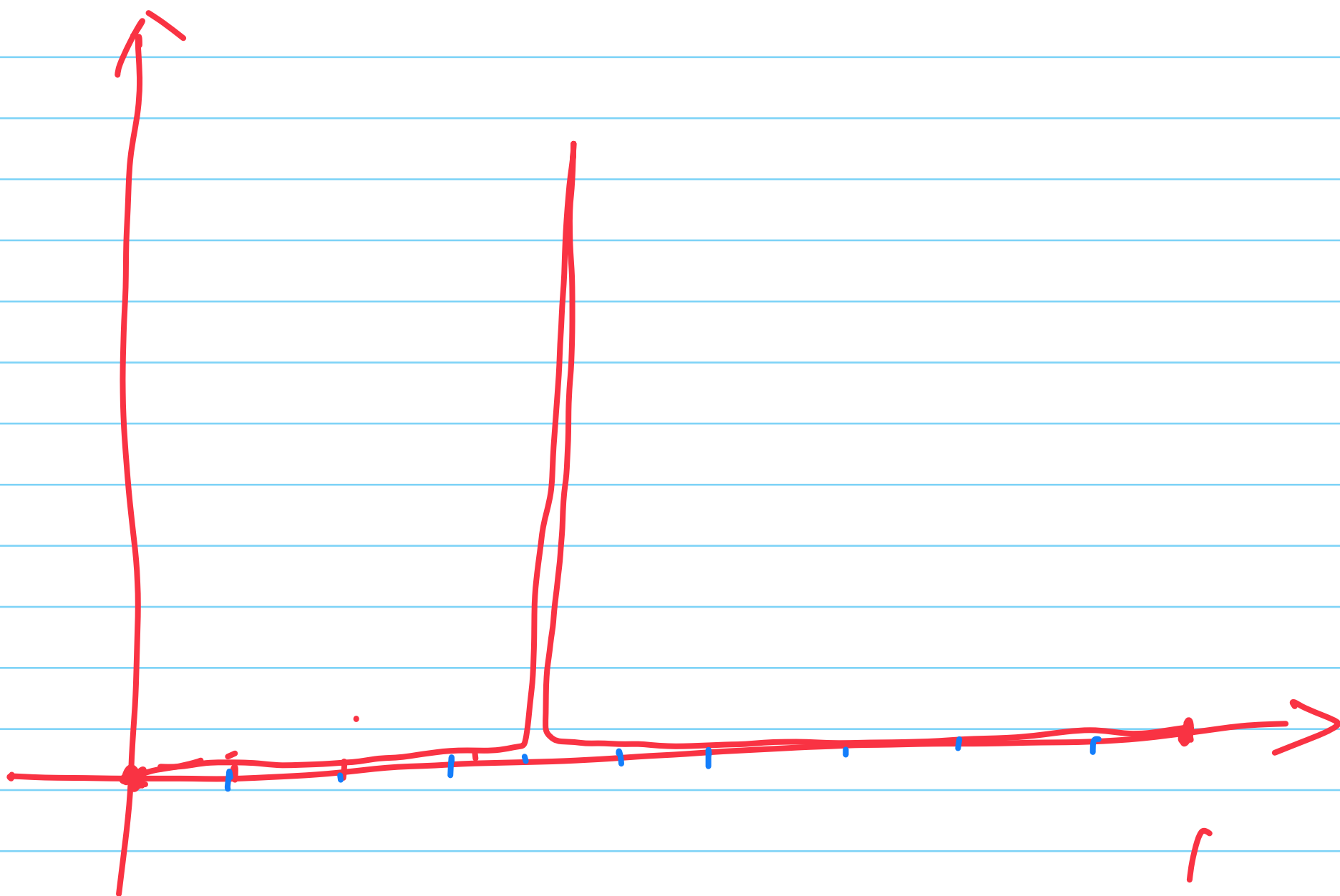
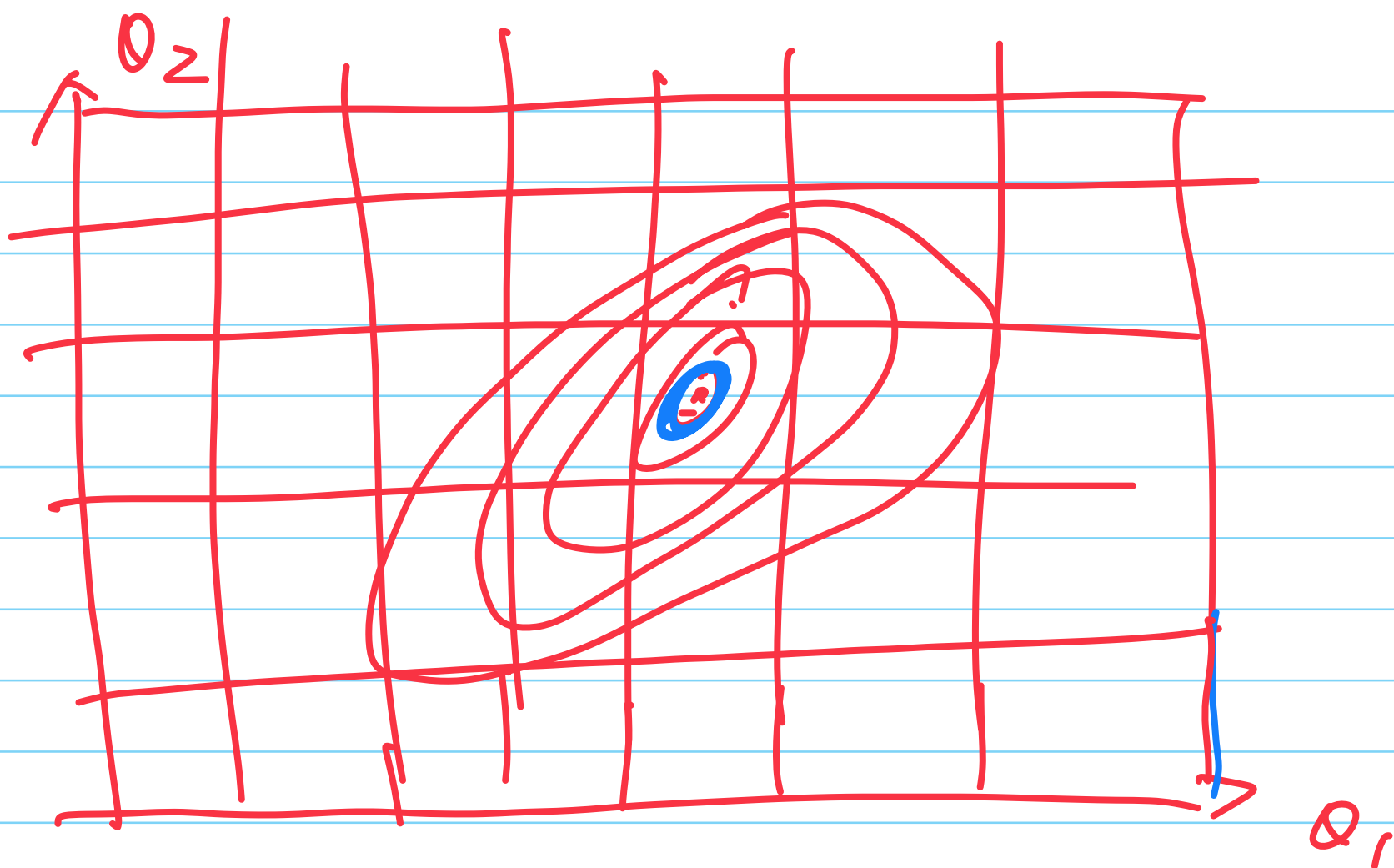
$$ML = \int L(\theta) \cdot \pi(\theta) d\theta$$



Difficulty of N. Q.

$$\int f(x) dx$$





Example:

